BEE 4750 Homework 3: Dissolved Oxygen and Monte Carlo

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Due Date

Thursday, 10/16/25, 9:00pm

Overview

Instructions

- Problem 1 asks you to implement a model for dissolved oxygen in a river with multiple waste releases and use this to develop a strategy to ensure regulatory compliance.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

```
  Activating project at `~/BEE 4750 /HW 3/hw3-parker`
Precompiling project...
    466.1 ms  WorkerUtilities
    472.1 ms  IteratorInterfaceExtensions
    471.7 ms  TensorCore
```

```
 536.2 ms    LaTeXStrings
 469.3 ms    Contour
 478.7 ms    StatsAPI
 356.6 ms    Measures
 918.2 ms    Format
 479.9 ms    Statistics
 503.9 ms    OrderedCollections
 653.6 ms    Grisu
 302.9 ms    DataValueInterfaces
 786.8 ms    FillArrays
 443.3 ms    Requires
 360.2 ms    StableRNGs
 443.3 ms    Unzip
 310.8 ms    Reexport
 502.0 ms    DocStringExtensions
 378.1 ms    SimpleBufferStream
 396.2 ms    PtrArrays
 612.3 ms    InlineStrings
 612.1 ms    URIs
1750.5 ms    MacroTools
 411.4 ms    InvertedIndices
1029.5 ms    IrrationalConstants
 499.4 ms    NaNMath
 475.3 ms    DelimitedFiles
 616.1 ms    TranscodingStreams
 421.0 ms    DataAPI
 355.2 ms    BitFlags
 376.4 ms    Scratch
 720.3 ms    ConcurrentUtilities
 982.5 ms    Crayons
 598.2 ms    LoggingExtras
 904.4 ms    MbedTLS
1071.9 ms    SentinelArrays
 754.2 ms    Preferences
 624.8 ms    Compat
 579.7 ms    ExceptionUnwrapping
1521.8 ms    PDMats
 613.5 ms    TableTraits
 704.5 ms    Statistics → SparseArraysExt
 401.9 ms    Showoff
1210.5 ms    UnicodeFun
 671.3 ms    FillArrays → FillArraysStatisticsExt
 637.4 ms    FillArrays → FillArraysSparseArraysExt
```

```
 567.4 ms   AliasTables
 791.2 ms   CodecZlib
1826.4 ms   FixedPointNumbers
1596.3 ms   DataStructures
 868.4 ms   Missings
 935.2 ms   PooledArrays
1450.9 ms   LogExpFunctions
 958.0 ms   RelocatableFolders
 750.2 ms   PrecompileTools
 767.0 ms   JLLWrappers
 886.9 ms   Compat → CompatLinearAlgebraExt
1105.7 ms   FillArrays → FillArraysPDMatsExt
1338.0 ms   SortingAlgorithms
1431.5 ms   QuadGK
1776.2 ms   ColorTypes
2331.3 ms   Tables
 742.6 ms   OpenSSL_jll
 529.5 ms   Graphite2_jll
 532.9 ms   Libmount_jll
2046.7 ms   RecipesBase
1927.2 ms   StringManipulation
 435.2 ms   EpollShim_jll
 412.7 ms   LLVMOpenMP_jll
 432.3 ms   Bzip2_jll
 391.7 ms   Xorg_libICE_jll
 430.6 ms   Rmath_jll
 397.4 ms   Xorg_libXau_jll
 416.3 ms   libfdk_aac_jll
 421.7 ms   libpng_jll
 438.8 ms   LAME_jll
 397.3 ms   fzf_jll
 406.7 ms   LERC_jll
 507.5 ms   JpegTurbo_jll
 472.8 ms   Ogg_jll
 508.2 ms   XZ_jll
 365.1 ms   mtdev_jll
 382.9 ms   Xorg_libXdmcp_jll
 419.3 ms   x265_jll
 455.0 ms   x264_jll
 441.8 ms   Zstd_jll
 455.9 ms   libaom_jll
 434.1 ms   Expat_jll
 424.8 ms   LZO_jll
```

```
 440.0 ms   Opus_jll
 403.7 ms   Xorg_xtrans_jll
 386.6 ms   libevdev_jll
 421.6 ms   Libiconv_jll
 414.1 ms   Libffi_jll
 412.5 ms   eudev_jll
 390.3 ms   Libuuid_jll
 658.7 ms   OpenSpecFun_jll
 416.4 ms   FriBidi_jll
 588.2 ms   FilePathsBase
 548.8 ms   ColorTypes → StyledStringsExt
1305.8 ms   Pixman_jll
2697.8 ms   ColorVectorSpace
 445.8 ms   Xorg_libSM_jll
1275.4 ms   FreeType2_jll
3301.6 ms   OpenSSL
1080.2 ms   Rmath
1350.9 ms   JLFzf
4577.8 ms   Colors
4931.8 ms   StatsBase
1324.6 ms   Ghostscript_jll
 720.4 ms   libvorbis_jll
 358.8 ms   Xorg_libxcb_jll
8681.3 ms   Parsers
 380.1 ms   Dbus_jll
 373.2 ms   Wayland_jll
 350.9 ms   libinput_jll
 876.7 ms   GettextRuntime_jll
 924.3 ms   FilePathsBase → FilePathsBaseMmapExt
1281.5 ms   FilePathsBase → FilePathsBaseTestExt
1828.1 ms   Libtiff_jll
 311.3 ms   Xorg_xcb_util_jll
1902.6 ms   SpecialFunctions
 307.1 ms   Xorg_libX11_jll
2611.8 ms   Fontconfig_jll
1728.4 ms   JSON
 316.3 ms   Xorg_xcb_util_image_jll
1792.6 ms   InlineStrings → ParsersExt
 363.8 ms   Xorg_xcb_util_keysyms_jll
 370.3 ms   Xorg_xcb_util_renderutil_jll
 441.5 ms   Xorg_xcb_util_wm_jll
2041.7 ms   Glib_jll
 509.0 ms   ColorVectorSpace → SpecialFunctionsExt
```

```
  411.5 ms    Xorg_libXrender_jll
 1155.3 ms    HypergeometricFunctions
 4498.8 ms    ColorSchemes
  389.4 ms    Xorg_libXext_jll
  474.1 ms    Xorg_libXfixes_jll
  422.3 ms    Xorg_libxkbfile_jll
  501.1 ms    Xorg_xcb_util_cursor_jll
  819.8 ms    WeakRefStrings
 4950.5 ms    Latexify
  453.8 ms    Xorg_libXinerama_jll
  400.1 ms    Xorg_libXrandr_jll
  376.6 ms    Libglvnd_jll
  360.8 ms    Xorg_libXi_jll
  368.8 ms    Xorg_libXcursor_jll
  407.4 ms    Xorg_xkbcomp_jll
 2207.4 ms    StatsFuns
 1296.5 ms    Latexify → SparseArraysExt
 1099.6 ms    Xorg_xkeyboard_config_jll
 2962.5 ms    Cairo_jll
  506.9 ms    xkbcommon_jll
  565.2 ms    Vulkan_Loader_jll
 1517.2 ms    HarfBuzz_jll
 1971.9 ms    libass_jll
  846.6 ms    Pango_jll
 5234.3 ms    Qt6Base_jll
 9496.1 ms    PlotUtils
  465.5 ms    libdecor_jll
 1608.7 ms    Qt6ShaderTools_jll
 3645.9 ms    FFMPEG_jll
 9047.8 ms    Distributions
  566.3 ms    GLFW_jll
 3991.1 ms    PlotThemes
 1767.3 ms    FFMPEG
 2572.4 ms    Qt6Declarative_jll
20109.4 ms    HTTP
  588.2 ms    Qt6Wayland_jll
 1508.2 ms    Distributions → DistributionsTestExt
 1629.5 ms    GR_jll
 4958.7 ms    RecipesPipeline
16889.5 ms    CSV
28853.5 ms    PrettyTables
 2785.7 ms    GR
29396.6 ms    DataFrames
```

```
   1600.2 ms    Latexify → DataFramesExt
  40598.2 ms    Plots
  177 dependencies successfully precompiled in 81 seconds. 37 already precompiled.
```

```julia
using Random
using Plots
using LaTeXStrings
using Distributions
```

Problems (Total: 30 Points)

Problem 1 (30 points)

A river which flows at 6 km/d is receiving waste discharges from two sources which are 15 km apart. The oxygen reaeration rate is 0.55 day-1, and the decay rates of CBOD and NBOD are are 0.35 and 0.25 day-1, respectively. The river's saturated dissolved oxygen concentration is 10m g/L.

If the characteristics of the river inflow and waste discharges are given in Table 1, write a Julia model to compute the dissolved oxygen concentration from the first wastewater discharge to an arbitrary distance `d` km downstream. Use your model to compute the minimum dissolved oxygen concentration up to 50 km downstream and how far downriver this maximum occurs.

Parameter	River Inflow	Waste Stream 1	Waste Stream 2
Inflow	100,000 m3/d	10,000 m3/d	15,000 m3/d
DO Concentration	7.5 mg/L	5 mg/L	5 mg/L
CBOD	5 mg/L	50 mg/L	45 mg/L
NBOD	5 mg/L	35 mg/L	35 mg/L

Table 1: River inflow and waste stream characteristics for Problem 1.

$K_a = 0.55$ per day

$K_c = 0.35$ per day

$K_n = 0.25$ per day

$C_s = 10$ mg/L

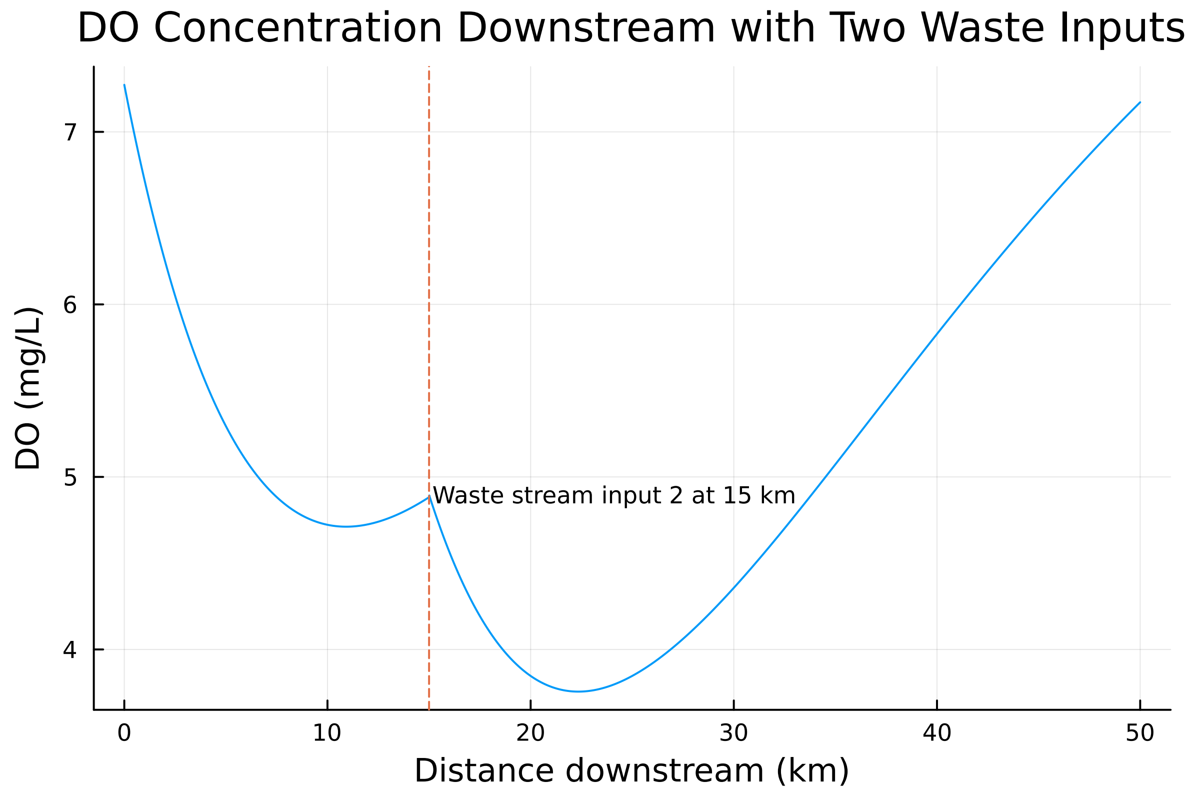
Problem 1.1

Implement the Streeter-Phelps (analytic) solution for the dissolved oxygen concentration. Plot the dissolved oxygen concentration from the first waste stream to 50 km downriver. What is the minimum value in mg/L?

```
u = 6 #km/day
dist = 15 #km
r = 0.55 #1/day
kc = 0.35 #1/day
kn = 0.25 #1/day
Cs = 10 #mg/L

Q_r0 = 100000 #m3/day
DO_r0 = 7.5 #mg/L
C_r0 = 5 #mg/L
N_r0 = 5 #mg/L

Q_w1 = 10000 #m3/day
DO_w1 = 5 #mg/L
C_w1 = 50 #mg/L
N_w1 = 35 #mg/L

Q_w2 = 15000 #m3/day
DO_w2 = 5 #mg/L
C_w2 = 45 #mg/L
N_w2 = 35 #mg/L

Q_out1 = Q_r0 + Q_w1

function conc_initial(c0, q0, c1, q1, qout)
    ((c0*q0) + (c1*q1)) / qout
end

DO_out1 = conc_initial(DO_r0, Q_r0, DO_w1, Q_w1, Q_out1)
C_out1  = conc_initial(C_r0,  Q_r0, C_w1,  Q_w1, Q_out1)
N_out1  = conc_initial(N_r0,  Q_r0, N_w1,  Q_w1, Q_out1)

function conc(x, u, r, kc, kn, DO0, C0, N0, Cs)
    a1 = exp(-r*x/u)
    a2 = (kc/(r-kc))*(exp(-kc*x/u) - a1)
    a3 = (kn/(r-kn))*(exp(-kn*x/u) - a1)
    Cs*(1 - a1) + DO0*a1 - C0*a2 - N0*a3
```

```julia
end

# UNTIL X = 15 RIGHT BEFORE SECOND WASTE STREAM INPUT
x1 = range(0.0, stop=15.0, length=1501)           # <- use 0..15 here
C1 = conc.(x1, u, r, kc, kn, DO_out1, C_out1, N_out1, Cs)

# values at x=15
# (L0 and N0 were undefined; you meant the mixed CBOD/NBOD after the first discharge)
L15 = C_out1 * exp(-kc*15.0/u)                    # scalar math is fine
N15 = N_out1 * exp(-kn*15.0/u)
C15 = C1[end]                                     # indexing uses [end], not C1(end)

# SECOND WASTE STREAM INPUT AT X = 15
Q_out2 = Q_out1 + Q_w2
DO_out2 = conc_initial(C15, Q_out1, DO_w2, Q_w2, Q_out2)
C_out2  = conc_initial(L15,  Q_out1, C_w2,  Q_w2, Q_out2)
N_out2  = conc_initial(N15,  Q_out1, N_w2,  Q_w2, Q_out2)

x2 = range(15.0, stop=50.0, length=3501)
C2 = conc.(x2 .- 15.0, u, r, kc, kn, DO_out2, C_out2, N_out2, Cs)

x_total = [x1; x2[2:end]]
C_total = [C1; C2[2:end]]
imin = argmin(C_total)
minDO = C_total[imin]
x_at_min = x_total[imin]
println("Minimum DO = $(round(minDO, digits=3)) mg/L at x = $(round(x_at_min, digits=3)) km")

# plot
plot(x_total, C_total, xlabel = "Distance downstream (km)", ylabel = "DO (mg/L)",
     title = "DO Concentration Downstream with Two Waste Inputs", legend = false)
vline!([15.0], linestyle = :dash)
annotate!(15.2, DO_out2, text("Waste stream input 2 at 15 km", :left, 8))
```

```
Minimum DO = 3.756 mg/L at x = 22.35 km
```



Problem 1.2

Implement a numerically-integrated (discretized) solution for the dissolved oxygen concentration. Using a resolution of 0.5km, conduct a simulation and plot the results on the same axis as the plot from Problem 1.1. How has the minimum value changed? What do you attribute this difference to (be specific about the source of the difference(s) in terms of the simulation dynamics, not just that one is a numerical approximation).

Problem 1.3

Using the analytic model, what is the minimum level of treatment (% removal of organic waste; assume this is equivalent to the same level of reduction of the CBOD and NBOD) for waste stream 1 that will ensure that the dissolved oxygen concentration is in compliance with the 4 mg/L standard along with a 5% margin of safety, assuming that waste stream 2 remains untreated? How about if only waste stream 2 is treated?

Problem 1.4

Suppose you are responsible for designing a waste treatment plan for discharges into the river, with a regulatory mandate to keep the dissolved oxygen concentration above 4 mg/L. Discuss whether you'd opt to treat waste stream 2 alone or both waste streams equally. What other information might you need to make a conclusion, if any?

Problem 1.5

Suppose that it is known that the DO concentrations at the river inflow can vary according to a $\text{LogNormal}(2.0, 0.15)$ distribution. Conduct a Monte Carlo simulation of the DO concentration in the river. If you treat only waste stream 1 (based on your analysis from Problem 1.3), what is the expected probability and 95% confidence interval that the river fails to comply with the regulatory standard of 4 mg/L? How did you decide that your Monte Carlo sample size was sufficiently large?

References

List any external references consulted, including classmates.